

Date : 15/12/2020

Max. Time: 50 minutes

Comprehensive Exam

Weightage: 17.5 %
Max. Marks: 35 Marks

Instructions to the students:

1. Consider the variables R and T as the last two digits of your ID number.
 If T is either 0 or 1 then consider $T=4$
 If R is either 0 or 1 then consider $R=6$
 (Examples:
 If the ID is 2017A3PS0589P, then T and R are 8 and 9, respectively
 If the ID is 2017A3PS0510P, then T and R are 4 and 6, respectively).
2. Clearly show the procedure used to arrive at your answers for full credit.
3. Write solution of each question on fresh page.
4. For every question write down the R and T values you are taking (it should match according to the instructions)
5. Each question carries equal weightage of marks.

Questions

1. I) Consider a multi-rate structure given below (Fig. 1A), where $H_0(z)$ and $H_1(z)$ are, respectively, ideal zero phase real coefficient low pass and high pass filters with the frequency responses as indicated in Fig. 1B. If the input is a real sequence whose DTFT as shown in Fig. 1C. Then sketch the DTFTs of the outputs $y_0[n]$ and $y_1[n]$

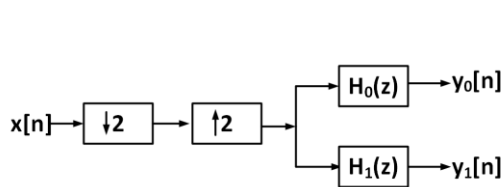


Fig. 1A

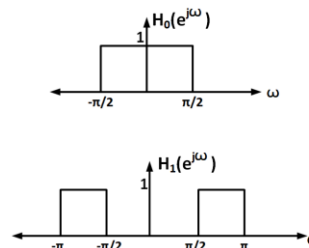


Fig. 1B

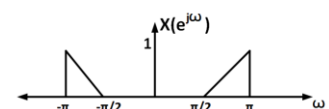


Fig. 1C

- II) Comment on whether an original signal can be recovered after the down-sampling operation. Justify your answer?
2. Consider a length-8 FIR transfer function given by
 $H(z) = R + Tz^{-1} + 2Rz^{-2} + 2Tz^{-3} + 2Tz^{-4} + 2Rz^{-5} + Tz^{-6} + Rz^{-7}$
 I) Draw the four-branch polyphase realization of $H(z)$. Write the expressions for the polyphase transfer functions $E_0(z)$, $E_1(z)$, $E_2(z)$, $E_3(z)$.
 II) Draw a canonical representation of the $H(z)$
3. A causal LTI system has an output $y[n] = (0.3)^n u[n] - 0.5(0.3)^{n-1} u[n-1]$, for an input $x[n] = (0.6)^n u[n]$. Then
 I) Draw a canonic direct-form realization of the system.
 II) Draw a parallel form-I realization of the system.

4. Design a Chebyshev Type-I digital IIR LPF using *bilinear transformation* by taking sampling period as 1 second to satisfy the following design specifications.

$$0.8 \leq |H(e^{j\omega})| \leq 1; \quad \text{for } 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2; \quad \text{for } 0.6\pi \leq \omega \leq \pi$$

5. I) A multi-rate structure is given below (Fig. 5). Determine the over-all transfer function $[W(z)=Y(z)/X(z)]$ in terms of $H_0(z)$ and $H_1(z)$ if:

$$P_0(z)=z^{-1}, P_1(z)=1+z^{-1}, Q_0(z)=1-z^{-1}, Q_1(z)=z^{-1}, \text{ and } Q_2(z)=z^{-2}-z^{-1}.$$

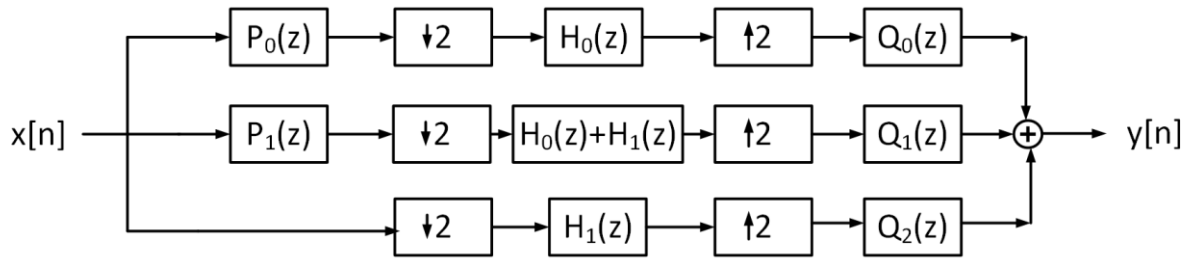


Fig. 5

- II) Determine the expression for $W(z)$ according to the equations given below. Then comment on the type of filter " $H(z)$ " if the multi-rate structure satisfies perfect reconstruction condition.

$$H_0(z^2) = \frac{H(z) + H(-z)}{2}, \quad H_1(z^2) = \frac{z[H(z) - H(-z)]}{2}$$

*** End of Paper ***